



# Evaluation of a Multi-robot Cafe Based on Service Quality Dimensions

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## Abstract

While AI applications are popular in many domains, they should work harmoniously with domain experts and end users. Furthermore, to develop such harmonious AI applications, we need agile AI platforms for not only developers, but also domain experts. Currently, we have developed PRactical INTElligent aPplicationS (PRINTEPS), which is a user-centric platform to develop integrated intelligent applications. This paper reports on a multi-robot cafe as a practical application of PRINTEPS and evaluates its service quality at a university festival. It is not clear if robotic services are perceived as attractive and/or valuable, and how the concept of robotic services could lead to customer satisfaction. Therefore, the evaluation of the quality of such services is necessary to identify the key factors that could contribute to improving customer satisfaction. The purpose of this research is to identify key factors in improving customer satisfaction in robotic services by evaluating the service quality of the multi-robot cafe. We designed questionnaire items based on SERVQUAL which is one of the service quality evaluation measurement methods and conducted a questionnaire survey at a multi-robot cafe held at a university festival. From the collected data, we modeled and evaluated the relationship between service quality and customer satisfaction to identify key factors in robotic services using a Bayesian network. In addition, the experiment confirms the usefulness of PRINTEPS.

**Keywords** PRINTEPS · multi-robot cafe · service quality · SERVQUAL

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## 1 Introduction

While AI applications are popular in many domains, they should work harmoniously with domain experts and end users. Furthermore, to develop such harmonious AI applications, we need agile AI platforms for not only developers, but also domain experts. Such AI platforms should have AI integration and agile process facilities. The former means combining four types of AI modules such as knowledge-based reasoning, speech dialog, image sensing, and motion management. The latter means an incremental AI apps development process to smartly accept domain expert's feedback.

To support the development of AI applications and/or service robot applications, several tools [1, 3, 13] and platforms [6, 15, 16, 19] have been proposed. However, many existing studies are for developers and/or researchers and tools for end users are limited in scope.

From the above background, we have developed PRACTICAL INTELLIGENT APPLICATIONS (PRINTEPS) [18], which is a user-centric platform to develop integrated intelligent applications by combining only four types of modules: knowledge-based reasoning, speech dialog, image sensing, and motion management. PRINTEPS supports end users in thoughtfully considering AI application design (user participation design) and in developing and revising such design easily.

This paper reports on a multi-robot cafe as a practical application of PRINTEPS and evaluates its service quality at a university festival. It is not clear if robotic services are perceived as attractive and/or valuable, and how the concept of robotic services could lead to customer satisfaction. Therefore, the evaluation of the quality of such services is necessary to identify the key factors that could contribute to improving customer satisfaction.

The purpose of this research is to identify the key factors in improving customer satisfaction in robotic services through evaluating the service quality of the multi-robot cafe. We designed questionnaire items based on SERVQUAL which is one of the service quality evaluation measurement methods and conducted a questionnaire survey at a multi-robot cafe held at a university festival. From the collected data, we modeled and evaluated the relationship between service quality and customer satisfaction to identify the key factors in robotic services. In addition, the experiment confirms the usefulness of PRINTEPS.

The remainder of this paper is organized as follows. In Sect. 2, we describe the setup environment and the service workflow of the multi-robot cafe. We also introduce the development support tools of PRINTEPS. In Sect. 3, we describe the design of service quality evaluation items for robotic services based on SERVQUAL. In addition, we introduce additional service quality evaluation items, entertainment factors, and interactivity. In Sect. 4, we describe evaluations of the multi-robot cafe based on the service quality evaluation items. We also describe a service quality evaluation model using a Bayesian network. In Sect. 5, we introduce related works, which we compare with our service quality evaluation model. Finally, we give our conclusions and future work in Sect. 6.

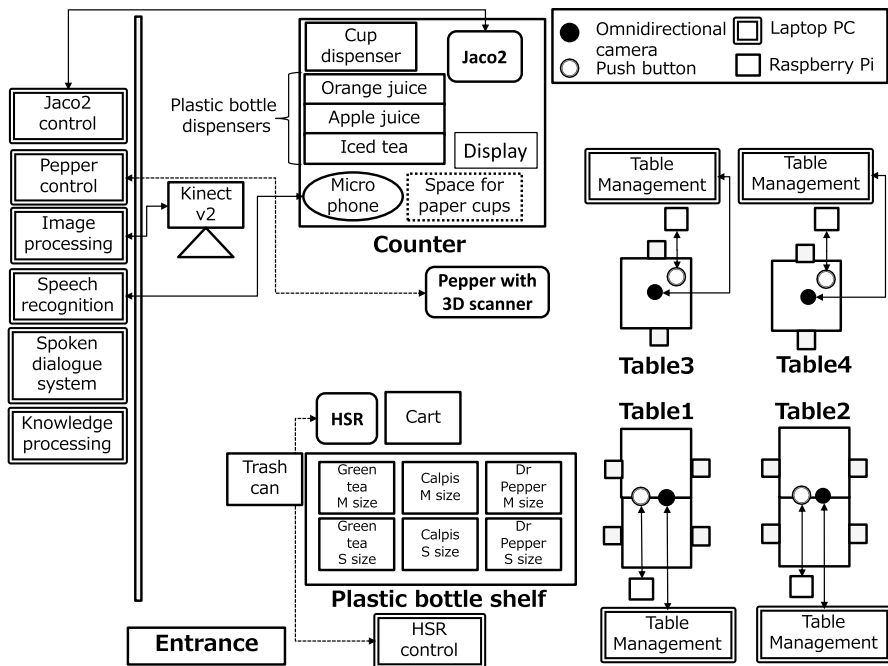


Fig. 1 Setup environment of the multi-robot cafe

## 2 Experiments of the Multi-robot Cafe

### 2.1 Setup Environment of the Multi-robot Cafe

The experiment was carried out at the students' lounge in the 34th building at Yagami campus of Keio University during the 18th Yagami Festival on Oct. 7 and 8, 2017. The setup environment of the multi-robot cafe is shown in Fig. 1. It shows the layout of the robot cafe in which the entrance, the service counter, two tables to seat four people each, two tables to seat two people each, plastic bottle dispensers, a cup dispenser, a plastic bottle shelf, a trash can, a cart, arm-type robot Jaco2, concierge robot Pepper, and serving robot HSR are depicted. Figure 2 shows an overview of the robot cafe at the students' lounge, Fig. 3 shows the counter of the robot cafe with Pepper and Jaco2, and Fig. 4 shows the plastic bottle shelf and the trash can of the robot cafe with HSR respectively.

Kinect v2, omnidirectional cameras, and each robot were connected to a control laptop PC through a wired (solid line in the figure) or a wireless (dotted line) connection. Push buttons placed on each table were connected to a Raspberry Pi. In addition, two other laptop PCs were installed; one was a knowledge processing PC to run the knowledge and inference process and the workflow created by PRINTEPS workflow editor; the other was for the spoken dialog system.



**Fig. 2** Overview of the robot cafe at the students' lounge

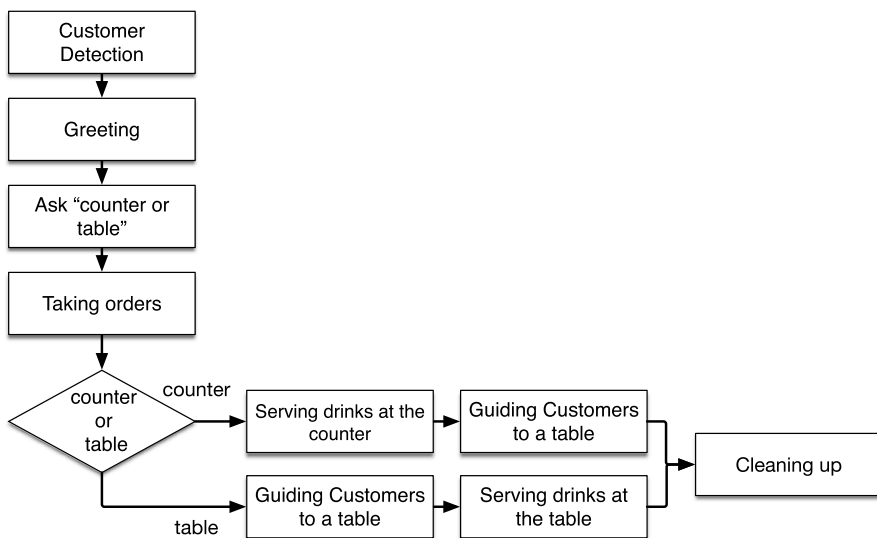


**Fig. 3** The counter of the robot cafe with Pepper and Jaco2

The multi-robot cafe, as explained above, had been designed and implemented using the PRINTEPS workflow editor. Please see [11] for details of the architecture of PRINTEPS and each elemental module.



**Fig. 4** HSR with the plastic bottle shelf and the trash can at the robot cafe



**Fig. 5** Service workflow in the robot cafe

## 2.2 Service Workflow in the Robot Cafe

The following section explains the service workflow in the robot cafe. Figure 5 shows the service workflow in the robot cafe.



**Fig. 6** Sequence of pictures showing Pepper and HSR moving without collision

### 2.2.1 Customer Detection, Greeting, and Order Taking

Kinect v2 was positioned in front of the entrance mainly for detecting and counting incoming guests. A microphone was placed on the counter for taking orders using speech recognition. First, when customers entered the robot cafe, Kinect v2 detected the entering store, and Pepper greeted the customers. Then, Pepper asked the customers whether to choose either counter service or table service. If the customers choose the counter service, the customers could order either orange juice, apple juice, iced tea, or mixed juice. If the customers choose the table service, the customers could order either Dr. Pepper, Calpis, or green tea.

### 2.2.2 Serving Drinks at the Counter

Three plastic bottle dispensers and a cup dispenser were placed on the counter to serve orange juice, apple juice, iced tea, and mixed juice. Jaco2 took out a paper cup from the cup dispenser according to the order, placed it below the corresponding plastic bottle dispenser, and pulled the lever for a duration corresponding to the size of the order to serve the drink to the guests. If mixed juice was ordered, equal quantities of both orange juice and apple juice were poured into the cup.

### 2.2.3 Guiding Customers to a Table

After taking the orders, Pepper-guided customers to a table available. Pepper had a 3D scanner to detect its own position during movement. In addition, Pepper could move while avoiding obstacles. Each table had an omnidirectional camera to detect people using the table to check if the table was unoccupied.

### 2.2.4 Serving Drinks at the Table

HSR took plastic bottle(s) from the plastic bottle shelf according to the orders and used the cart to carry the bottles when more than one bottle was ordered. Otherwise, it directly brought the bottle(s) to the table. HSR also had a 3D scanner to detect its own position during movement. Figure 6 shows the sequence of

**Table 1** Main ROS topics of the multi-robot cafe

| ROS topic names (Output type)                            | Description                                                             |
|----------------------------------------------------------|-------------------------------------------------------------------------|
| /printeps/cafe/visiting_customers_sensor (std_msgs/Bool) | Topics for detection of incoming guests                                 |
| /printeps/cafe/entrance (printeps_cafe_msgs/Entrance)    | Topics for the people around the entrance                               |
| /printeps/cafe/table1 (std_msgs/Bool)                    | Status of table 1 (empty or not)                                        |
| /printeps/cafe/counter/listener (std_msgs/String)        | Topics for speech recognition results with a microphone at the entrance |
| /pepper01/talker (std_msgs/String)                       | Topics for completion of orders                                         |
| /clean_up (std_msgs/String)                              | Topics for disposal of plastic bottles by HSR                           |

pictures showing Pepper and HSR moving without collision using the 3D scanner. S-size and M-size plastic bottles of Dr Pepper, Calpis, and green tea were placed on the bottle shelf, as shown in Fig. 1.

### 2.2.5 Cleaning Up Empty Plastic Bottles or Cups

Tables were also equipped with push buttons to call HSR, so that the robot would come to the table and collect empty plastic bottles. Guests were supposed to hand the empty bottles to the robot.

## 2.3 Development Support Tools of PRINTEPS

To set up a multi-robot cafe based on PRINTEPS, data from various sensors and devices, such as a microphone, Kinect v2, and omnidirectional cameras, must be published as Robot Operating System (ROS) ([www.ros.org](http://www.ros.org)) topics. Table 1 shows the main ROS topics of the multi-robot cafe. Figure 7 shows the relationships among devices/sensors, ROS topics, and PRINTEPS modules/processes.

It is costly to create a program that publishes topics using ROS libraries by preparing the ROS development environment for more than one PC. In this experiment, the development support tool of PRINTEPS, shown in Fig. 8, was implemented. This tool was implemented using rosjava ([github.com/rosjava](https://github.com/rosjava)). It can execute on an OS on which the Java virtual machine runs and can publish ROS topics from yaml format file conforming to the ROS message. Therefore, development skills for ROS are not necessary for publishing ROS topics as long as the function to output the processing results of image sensors, etc., in yaml format can be implemented. For example, the ROS topic of the status of Table 1 can be published as topic “/printeps/cafe/table1” using this tool if a program is created to output “data: true” or “data: false” to a file. For this experiment, this tool was used to publish ROS topics regarding the status of the tables, speech recognition results, and the detection results of incoming guests (modules depicted as bolded round rectangles in Fig. 7).

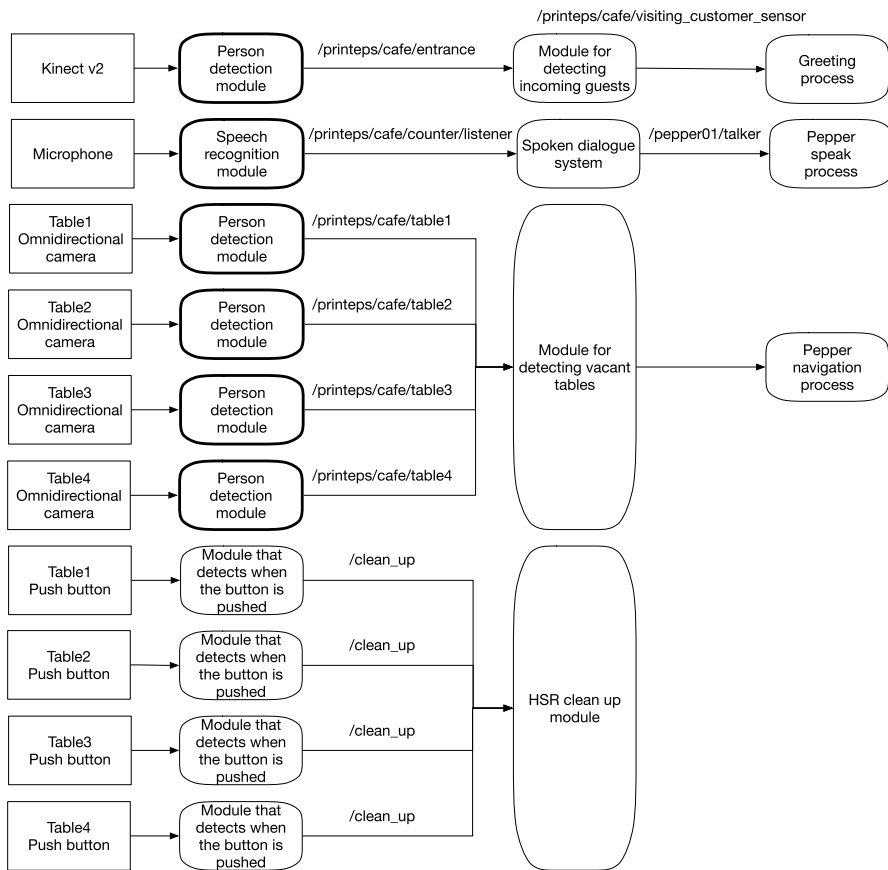


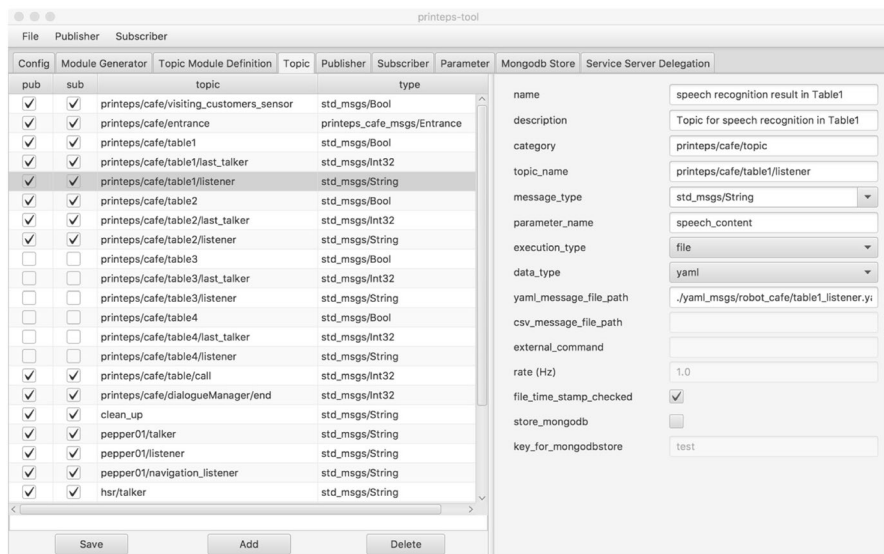
Fig. 7 Relationships among devices/sensors, ROS topics, and PRINTEPS modules/processes

### 3 Design of Service Quality Evaluation Items for Robotic Services

#### 3.1 Service Quality Evaluation Items Based on SERVQUAL

Parasuraman et al. [12] proposed a service quality evaluation method called SERVQUAL to access customer perceptions of service quality in service and retailing organizations. SERVQUAL consists of five dimensions of the service experience. Table 2 shows SERVQUAL's five dimensions.

In [12], a 22-item instrument was developed, and each item was categorized as one of the five dimensions. The 22-item instrument is shown below. In the following list, items 1–4 are categorized as tangibles, items 5–9 are categorized as reliability, items 10–13 are categorized as responsiveness, items 14–17 are categorized as assurance, and items 18–22 are categorized as empathy.



**Fig. 8** Development support tools of PRINTEPS

**Table 2** SERVQUAL's five dimensions

|                |                                                                                       |
|----------------|---------------------------------------------------------------------------------------|
| Assurance      | Knowledge and courtesy of employees and their ability to inspire trust and confidence |
| Empathy        | Caring, individualized attention the firm provides its customers                      |
| Reliability    | Ability to perform the promised service dependably and accurately                     |
| Responsiveness | Willingness to help customers and provide prompt service                              |
| Tangibles      | Physical facilities, equipment, and appearance of personnel                           |

## Tangibles

1. XYZ has up-to-date equipment.
2. XYZ's physical facilities are visually appealing.
3. XYZ's employees are well dressed and appear neat.
4. The appearance of the physical facilities of XYZ is in keeping with the type of services provided.

## Reliability

5. When XYZ promises to do something by a certain time, it does so.
6. When you have problems, XYZ is sympathetic and reassuring.
7. XYZ is dependable.
8. XYZ provides its services at time it promises to do so.
9. XYZ keeps its records accurately.

## Responsiveness

10. XYZ does not tell customers exactly when services will be performed.
11. You do not receive prompt service from XYZ's employees.
12. Employees of XYZ are not always willing to help customers.
13. Employees of XYZ are too busy to respond to customer requests promptly.

## Assurance

14. You can trust employees of XYZ.
15. You feel safe in your transactions with XYZ's employees.
16. Employees of XYZ are polite.
17. Employees get adequate support from XYZ to do their jobs well.

## Empathy

18. XYZ does not give you individual attention.
19. Employees of XYZ do not give you personal attention.
20. Employees of XYZ do not know what your needs are.
21. XYZ does have your best interests at hearts.
22. XYZ does not have operating hours convenient to all their customers.

The procedure to evaluate the service quality of a target company based on SERVQUAL is as follows. First, ask customers to rate all these items by their degree of expectation for the service of the target company. After that, ask the customers to evaluate all items by their perceived quality after service use. Finally, a person who wants to know service quality of the target company takes the difference between the perceived quality and the expectation among the items as the measurement value of the service quality.

Depending on the targeted service, expectations may not be clearly formed, and measurement may be difficult. In such cases, Cronin and Taylor proposed a method to measure the evaluation items only with perceptual quality. This method is called SERVPERF [2]. The robot cafe service targeted by this research is not widely spread in the world and evaluating of the degree of expectation before experiencing the service is difficult. Therefore, in this research, we used the SERVPERF approach to evaluate the service quality of the robot cafe.

We have developed 11 service quality evaluation items based on SERVQUAL's five dimensions and the 22-item instrument for the robot cafe (see Table 3).

## 3.2 Additional Service Quality Evaluation Items for Robotic Services

SERVQUAL has been applied to various service fields so far [9]. The five dimensions are concepts that cover various aspects of the services and these dimensions are important in many service fields. However, SERVQUAL is an index to measure the quality of services offered by people. We believe that the five dimensions are not

**Table 3** Evaluation items based on SERVQUAL regarding service quality

| ID  | Item                                                         | Type           |
|-----|--------------------------------------------------------------|----------------|
| S1  | The robots can provide what customers ordered.               | Reliability    |
| S2  | The robots can carry out services properly.                  | Reliability    |
| S3  | The time required by the robots to serve is appropriate      | Responsiveness |
| S4  | Customers feel at ease using the robot cafe.                 | Assurance      |
| S5  | The actions of the robots give customers a feeling of trust. | Assurance      |
| S6  | The robots are polite to customers.                          | Assurance      |
| S7  | The robots have sufficient knowledge about services          | Assurance      |
| S8  | Some of the robots pay attention to the customers.           | Empathy        |
| S9  | The robot gestures and attitudes are neat                    | Tangibles      |
| S10 | The appearance of the robots is good.                        | Tangibles      |
| S11 | The robots have an appearance that is suitable for a cafe    | Tangibles      |

**Table 4** Evaluation items regarding the interactivity and entertainment factor of the robots

| ID | Item                                                                                                                     | Type                 |
|----|--------------------------------------------------------------------------------------------------------------------------|----------------------|
| I1 | The robots understand the customer's conversation.                                                                       | Interactivity        |
| I2 | Customers can easily call the robots when they want something                                                            | Interactivity        |
| E1 | Customers feel attached to the robots.                                                                                   | Entertainment factor |
| E2 | Conversation with the robots is enjoyable.                                                                               | Entertainment factor |
| E3 | Watching the movement of the robots is interesting.                                                                      | Entertainment factor |
| E4 | It is cozy being in the robot cafe.                                                                                      | Entertainment factor |
| E5 | Robotic service is an innovative idea.                                                                                   | Entertainment factor |
| E6 | I am moved by the appearance of<br>Pepper and HSR moving without collision.                                              | Entertainment factor |
| E7 | I am surprised that the robot will guide me a seat available.                                                            | Entertainment factor |
| E8 | It is interesting that the robot will clean up<br>plastic bottles and paper cups<br>by pressing the button on the table. | Entertainment factor |

adequate enough to evaluate the “services provided by the robots” to be studied in this research.

Therefore, we have developed and an “interactivity (conversation) of the robots” and an “entertainment factor of the robots” in addition to the five dimensions. Table 4 shows the related items in each additional dimension.

The “Interactivity of the robots” is a dimension for evaluating how robots interact with people. We believe that customer satisfaction cannot be improved with the robots working as programmed. The robots can read, talk, and consider the customers intentions; therefore, the authors consider the robots to provide a high level of customer satisfaction.

The “Entertainment factor of the robots” is a dimension for evaluating how robots amuse people. We consider that robots doing work that human beings can take for

**Table 5** General evaluation items of the robot cafe

| ID | Item                                                                                        | Type                                            |
|----|---------------------------------------------------------------------------------------------|-------------------------------------------------|
| G1 | Generally satisfied with the service of the robot cafe.                                     | Customer satisfaction                           |
| G2 | Service of the robot cafe is up to my expectation.                                          | Customer satisfaction                           |
| G3 | Would like to use the robot cafe again.                                                     | Intention to revisit the robot cafe             |
| G4 | Would like to introduce the robot cafe to my family members, friends, and/or acquaintances. | Intention to recommend the robot cafe to others |

granted will also contribute to entertaining people. For example, we developed an item of “waiting for the movement of the robot is interesting”.

### 3.3 Service Quality Evaluation Items for Customer Satisfaction and Customer Loyalty

Customer satisfaction with service can be measured by asking the customer about the degree of overall satisfaction of the service. Therefore, we have developed two items that ask customers about their general satisfaction with the multi-robot cafe to measure customer satisfaction (see G1 and G2 in Table 5).

Recently, the concept of customer loyalty has been emphasized, and Reichheld, for example, argues that customer loyalty is very strongly related to management results, and if loyalty is high, it will lead to growth of the company [14]. Customer loyalty represents the customer’s will that customers desire to purchase and to continuously receive products and services from the company. By asking “intention to reuse” and “intention to recommend to others”, customer loyalty can be measured. Therefore, we also have developed two items related to customer loyalty (see G3 and G4 in Table 5).

## 4 Evaluation

### 4.1 Attributes of the Respondents

The questionnaire survey was administered to the guests of the multi-robot cafe at the 18th Yagami Festival of Keio University on Oct. 7 and 8, 2017. Questionnaires were handed to the guests to be filled out after the service. The sheets were retrieved on the spot. Table 6 shows the attributes of the respondents.

There were 95 effective respondents in the 18th Festival. Attributes of the respondents were: gender, 49.5% male and 50.5% female; age 11.7% less than 10 years, 14.9% from 10 to 20 years old, 17.0% in their 20s, 14.9% 30s, 24.5% 40s, and 17.0% 50s; and accompanying person, 2.1% alone, 22.3% with friends, and 75.5% with family members. As can be seen here, the majority of the guests were family members.

**Table 6** The attributes of the respondents

|                     |                        |       |
|---------------------|------------------------|-------|
| Gender              | Male                   | 49.5% |
|                     | Female                 | 50.5% |
| Age                 | Less than 10 years old | 11.7% |
|                     | 10–19 years old        | 14.9% |
|                     | 20–29 years old        | 17.0% |
|                     | 30–39 years old        | 14.9% |
|                     | 40–49 years old        | 24.5% |
|                     | 50–59 years old        | 17.0% |
|                     | More than 60 years old | 0.0%  |
| Accompanying person | Alone                  | 2.1%  |
|                     | Friends                | 22.3% |
|                     | Family members         | 75.5% |

**Table 7** Service quality evaluation of the multi-robot cafe ranked by average values

| Ranking | Item ID | Type                 | Average | Standard deviation |
|---------|---------|----------------------|---------|--------------------|
| 1       | E3      | Entertainment factor | 6.35    | 1.029              |
| 2       | S1      | Reliability          | 6.32    | 1.265              |
| 3       | E5      | Entertainment factor | 6.20    | 1.258              |
| 4       | E8      | Entertainment factor | 6.19    | 1.037              |
| 5       | S6      | Assurance            | 6.15    | 1.229              |
| 6       | S10     | Tangibles            | 5.87    | 1.196              |
| 7       | E1      | Entertainment factor | 5.84    | 1.331              |
| 8       | E2      | Entertainment factor | 5.77    | 1.440              |
| 9       | S9      | Tangibles            | 5.64    | 1.263              |
| 10      | S2      | Reliability          | 5.38    | 1.696              |
| 11      | E4      | Entertainment factor | 5.31    | 1.321              |
| 12      | E6      | Entertainment factor | 5.19    | 1.664              |
| 13      | E7      | Entertainment factor | 5.17    | 1.752              |
| 14      | I1      | Interactivity        | 4.97    | 1.691              |
| 15      | S11     | Tangibles            | 4.84    | 1.582              |
| 16      | S7      | Assurance            | 4.83    | 1.389              |
| 17      | S4      | Assurance            | 4.76    | 1.668              |
| 18      | S5      | Assurance            | 4.63    | 1.571              |
| 19      | I2      | Interactivity        | 4.40    | 1.569              |
| 20      | S8      | Empathy              | 3.88    | 1.703              |
| 21      | S3      | Responsiveness       | 3.50    | 1.664              |

## 4.2 Evaluation of Service Quality, Customer Satisfaction, and Customer Loyalty

This section describes the evaluation results of the multi-robot cafe at the 18th Yagami Festival of Keio University.

**Table 8** Average scores for each of the seven dimensions

| Dimension            | Average |
|----------------------|---------|
| Reliability          | 5.85    |
| Entertainment factor | 5.75    |
| Tangibles            | 5.45    |
| Assurance            | 5.09    |
| Interactivity        | 4.69    |
| Empathy              | 3.88    |
| Responsiveness       | 3.50    |

The evaluation scale of the questionnaire items shown in Tables 3, 4, and 5 was from “1: Not at all” through “7: Very much so.”

Table 7 shows the service quality evaluation of the multi-robot cafe ranked by average values. Table 8 shows the average scores for each of the seven dimensions. Each Item ID in Table 7 refers to the ID in Tables 3 and 4. These results indicate that the evaluation of the entertainment factor, reliability, and the tangibles of the services are high, while responsiveness, empathy, and interactivity of the services are low.

In this robot cafe, when the customer ordered from the robot, it was necessary to talk in a form according to prepared templates. In addition, Pepper was the only interactive robot. Therefore, the evaluation of empathy and interactivity was low.

Regarding the assurance, the item evaluating politeness (S6 in Table 7) was 6.15. However, the items evaluating the robot’s knowledge and the ability to give trust and ease to customers (S4, S5, and S7 in Table 7) were lower than 5.0. This robot cafe was at the experimental stage and unexpected difficulties occurred many times. We assume that these were some of the reasons why the evaluations of the latter items related to the assurance were low.

Regarding the entertainment factor, it is noteworthy that evaluation of the item about cleaning up plastic bottles and cups was 6.19 (E8 in Table 7), whereas the evaluation of the item about guiding customers to a seat available and the robots moving without collision was 5.17 and 5.19, respectively (E7 and E6 in Table 7). From these results, the robot moving without colliding and guiding the customers to a seat available was not recognized as “performance of the robots” by some of the customers. Therefore, to improve the evaluation of the entertainment factor, the customers have to be informed that these items are part of the performance of the robots.

Table 9 shows a general evaluation of the multi-robot cafe ranked by average values. Each Item ID in Table 9 refers to the ID in Table 5.

It is noteworthy that the evaluation of the “Intention to recommend the robot cafe to others” (5.76) is higher than the evaluation of the “Intention to revisit the robot cafe” (5.18). Since the robot cafe is a new service, new customers are easy to visit. However, according to Table 9, it seems that some customers do not want to return to the robot cafe again. Therefore, to acquire repeat customers and to

**Table 9** General evaluation of the multi-robot cafe ranked by average values

| Ranking | Item ID | Type                                            | Average | Standard deviation |
|---------|---------|-------------------------------------------------|---------|--------------------|
| 1       | G4      | Intention to recommend the robot cafe to others | 5.76    | 1.397              |
| 2       | G1      | Customer satisfaction                           | 5.18    | 1.406              |
| 3       | G3      | Intention to revisit the robot cafe             | 5.18    | 1.608              |
| 4       | G2      | Customer satisfaction                           | 4.66    | 1.542              |

continue the service, maintaining entertainment and improving the quality of the other six dimensions are necessary.

### 4.3 Evaluation of the Free Description Questionnaire

We also conducted a free description questionnaire to the customers to find out why the customers were generally satisfied or not satisfied with our robot cafe. The following is a summary from the free description questionnaire by the customers who were generally satisfied with our robot cafe.

- It was fun to experience a future coffee shop.
- It was surprising that the robots provided services like human beings.
- Jaco2 was cool.

The positive opinions were mainly related to the entertainment factor.

The following is a summary from the free description questionnaire by the customers who were not generally satisfied with our robot cafe.

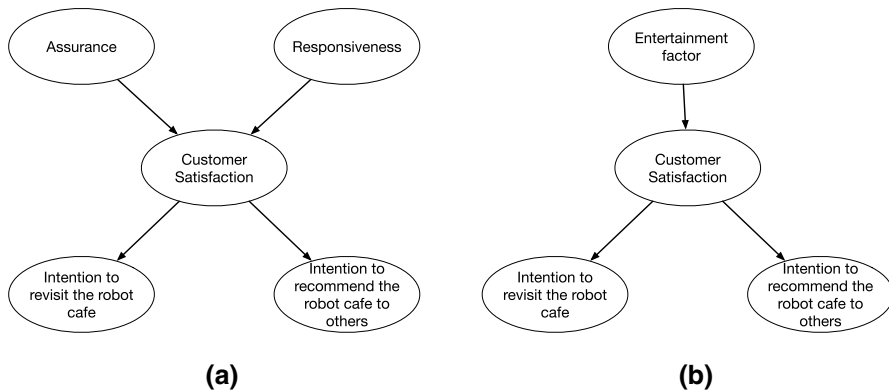
- The robots' motion and reaction were slow.
- It was a bit dangerous and the drink was going to spill.

The negative opinions were mainly related to the responsiveness and the assurance items.

### 4.4 Evaluation of Service Quality Evaluation Model

#### 4.4.1 Construction of Service Quality Evaluation Model

To model the relationship between service quality and customer satisfaction, we constructed a service quality evaluation model using a Bayesian network. A Bayesian network is a probabilistic graphical model having nodes of random subject



**Fig. 9** Service quality evaluation models of the multi-robot cafe

variables and dependencies of the nodes. We assumed that the seven variables of the service quality were independent from each other, and that these variables were specified as the candidates of the parent nodes of “General satisfaction,” “Intention to revisit the robot cafe,” and “Intention to recommend the robot cafe to others” to construct the model. Analysis software BayoNet<sup>1</sup> was used for the analysis.

The evaluation results of the service quality were averaged for each dimension to obtain variables of the modeling. There are seven dimensions (types) of the “5 dimensions of SERVQUAL,” “interactivity,” and “entertainment factor.” The variables have three values, i.e., “1: low” corresponding to the evaluation scale of 1 to less than 4, “2: medium” corresponding to the evaluation scale of 4 to less than 6, and “3: high” corresponding to the evaluation scale of 6 or more. General evaluation results were also converted from 7 value data to 3 value data.

Since respondents generally tend to rate items highly in the survey, we skewed the evaluation scale when converting it from 7 values into 3 values. Actually in our survey, the average rate of each value are 1: 3.3%, 2: 4.7%, 3: 5.5%, 4: 16.8%, 5: 18.0 %, 6: 22.1%, 7: 29.5%, respectively. Therefore, we set the range of low as broader than the range of high in this study.

#### 4.4.2 Service Quality Evaluation Models

The service quality evaluation models obtained from the analysis results are shown in Fig. 9. Figure 9a shows a model when five dimensions of SERVQUAL are set as explanatory variables. Figure 9b shows a model when two additional dimensions (interactivity and entertainment factor) are set as explanatory variables.

The Conditional Probability Table (CPT) of each objective variable is shown in Tables 10, 11, 12, 13, 14, and 15. Tables 16, 17, 18, 19, and 20 show posterior probabilities of “object variable value = 3” when the state of one node is fixed to 3.

<sup>1</sup> <http://www.msi.co.jp/bayolink/>

**Table 10** CPT of the “customer satisfaction” node in Fig. 9a

| Parent node               |           | Customer satisfaction |       |       |
|---------------------------|-----------|-----------------------|-------|-------|
| Variable                  | Condition | 1                     | 2     | 3     |
| Assurance, Responsiveness | 1,1       | 0.308                 | 0.539 | 0.154 |
|                           | 1,2       | 0.600                 | 0.200 | 0.200 |
|                           | 1,3       | 0.333                 | 0.333 | 0.333 |
|                           | 2,1       | 0.139                 | 0.611 | 0.250 |
|                           | 2,2       | 0.040                 | 0.440 | 0.520 |
|                           | 2,3       | 0.250                 | 0.250 | 0.500 |
|                           | 3,1       | 0.200                 | 0.200 | 0.600 |
|                           | 3,2       | 0.100                 | 0.200 | 0.700 |
|                           | 3,3       | 0.083                 | 0.083 | 0.833 |

**Table 11** CPT of “Intention to revisit the robot cafe” node in Fig. 9a

| Parent node           |           | Intention to revisit the robot cafe |       |       |
|-----------------------|-----------|-------------------------------------|-------|-------|
| Variable              | Condition | 1                                   | 2     | 3     |
| Customer satisfaction | 1         | 0.462                               | 0.308 | 0.231 |
|                       | 2         | 0.195                               | 0.610 | 0.195 |
|                       | 3         | 0.067                               | 0.156 | 0.778 |

**Table 12** CPT of “Intention to recommend the robot cafe to others” node in Fig. 9a

| Parent node           |           | Intention to recommend the robot cafe to others |       |       |
|-----------------------|-----------|-------------------------------------------------|-------|-------|
| Variable              | Condition | 1                                               | 2     | 3     |
| Customer satisfaction | 1         | 0.385                                           | 0.231 | 0.385 |
|                       | 2         | 0.048                                           | 0.476 | 0.476 |
|                       | 3         | 0.022                                           | 0.067 | 0.911 |

**Table 13** CPT of “customer satisfaction” node in Fig. 9b

| Parent node          |           | Customer satisfaction |       |       |
|----------------------|-----------|-----------------------|-------|-------|
| Variable             | Condition | 1                     | 2     | 3     |
| Entertainment factor | 1         | 0.571                 | 0.286 | 0.143 |
|                      | 2         | 0.143                 | 0.592 | 0.265 |
|                      | 3         | 0.044                 | 0.261 | 0.696 |

#### 4.4.3 Discussion

From Fig. 9a, “assurance” and “responsiveness” are related to “customer satisfaction”. Regarding customer loyalty, both the “intention to revisit the robot cafe” and

**Table 14** CPT of “Intention to revisit the robot cafe” node in Fig. 9b

| Parent node           |           | Intention to revisit the robot cafe |       |       |
|-----------------------|-----------|-------------------------------------|-------|-------|
| Variable              | Condition | 1                                   | 2     | 3     |
| Customer satisfaction | 1         | 0.462                               | 0.308 | 0.231 |
|                       | 2         | 0.191                               | 0.619 | 0.191 |
|                       | 3         | 0.065                               | 0.152 | 0.783 |

**Table 15** CPT of “Intention to recommend the robot cafe to others” node in Fig. 9b

| Parent node           |           | Intention to recommend the robot cafe to others |       |       |
|-----------------------|-----------|-------------------------------------------------|-------|-------|
| Variable              | Condition | 1                                               | 2     | 3     |
| Customer satisfaction | 1         | 0.385                                           | 0.231 | 0.385 |
|                       | 2         | 0.047                                           | 0.465 | 0.488 |
|                       | 3         | 0.022                                           | 0.065 | 0.913 |

**Table 16** Posterior probability of “customer satisfaction” in Fig. 9a

| Prior probability with “customer satisfaction” = 3 is 0.422 |                                                        |
|-------------------------------------------------------------|--------------------------------------------------------|
| Instrumental variable                                       | Posterior probability with “customer satisfaction” = 3 |
| Assurance                                                   | 0.665                                                  |
| Responsiveness                                              | 0.560                                                  |

**Table 17** Posterior probability of “Intention to revisit the robot cafe” in Fig. 9a

| Prior probability with “Intention to revisit the robot cafe” = 3 is 0.447 |                                                                      |
|---------------------------------------------------------------------------|----------------------------------------------------------------------|
| Instrumental variable                                                     | Posterior probability with “Intention to revisit the robot cafe” = 3 |
| Customer satisfaction                                                     | 0.778                                                                |
| Assurance                                                                 | 0.588                                                                |
| Responsiveness                                                            | 0.529                                                                |

“intention to recommend the robot cafe to others” are combined only with “customer satisfaction”. In other words, when the state of “customer satisfaction” is decided, they become conditionally independent with other nodes.

From Fig. 9b, “entertainment factor” is related to “customer satisfaction”. Regarding customer loyalty, both the “intention to revisit the robot cafe” and “intention to recommend the robot cafe to others” are combined only with “customer satisfaction”.

**Table 18** Posterior probability of “Intention to recommend the robot cafe to others” in Fig. 9a

| Prior probability with “Intention to recommend the robot cafe to others” = 3 is 0.644 |                                                                                  |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Instrumental variable                                                                 | Posterior probability with “Intention to recommend the robot cafe to others” = 3 |
| Customer satisfaction                                                                 | 0.911                                                                            |
| Assurance                                                                             | 0.752                                                                            |
| Responsiveness                                                                        | 0.699                                                                            |

**Table 19** Posterior probability of “Intention to revisit the robot cafe” in Fig. 9b

| Prior probability with “Intention to revisit the robot cafe” = 3 is 0.465 |                                                                      |
|---------------------------------------------------------------------------|----------------------------------------------------------------------|
| Instrumental variable                                                     | Posterior probability with “Intention to revisit the robot cafe” = 3 |
| Customer satisfaction                                                     | 0.783                                                                |
| Entertainment factor                                                      | 0.604                                                                |

**Table 20** Posterior probability of “Intention to recommend the robot cafe to others” in Fig. 9b

| Prior probability with “Intention to recommend the robot cafe to others” = 3 is 0.669 |                                                                                  |
|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Instrumental variable                                                                 | Posterior probability with “Intention to recommend the robot cafe to others” = 3 |
| Customer satisfaction                                                                 | 0.913                                                                            |
| Entertainment factor                                                                  | 0.779                                                                            |

From Tables 13 and 16, “entertainment factor”, “assurance”, “responsiveness” can be factors for increasing the posterior probability of “customer satisfaction”. In particular, “entertainment factor” and “assurance” are important factors for “customer satisfaction”.

From Tables 17 and 19, “customer satisfaction”, “entertainment factor”, and “assurance” can be factors for increasing the posterior probability of “intention to revisit the robot cafe”.

From Tables 18 and 20, “customer satisfaction”, “entertainment factor”, and “assurance” can be factors for increasing the posterior probability of “intention to recommend the robot cafe to others”.

From the above results, “entertainment factor” is an important factor in any objective variable and it is an indispensable factor in realizing the robot cafe service.

In terms of customer satisfaction, “assurance” and “responsiveness” are also key factors. In other words, to improve customer satisfaction, it is considered necessary to improve the service speed of the robot and enhance the ability to carefully carry out the promised service without delay.

“Customer satisfaction” is also important for the “intention to revisit the robot cafe” and the “intention to recommend the robot cafe to others”, and it strongly influences customers. Improving customer satisfaction is effective in increasing repeaters and new customers.

## 5 Related Work

[8] proposes a theoretical model to explain the adoption process of the Ubiquitous Robot Companion (URC) personal robot service. A total of 30 measurement items were developed for tangibles, responsiveness, assurance, system quality, perceived usefulness, user satisfaction, and intention to use on the basis of the service quality and IS (Information Systems) research literatures [4], in an effort to characterize the quality, satisfaction, and adoption of the personal robot service. 490 respondents collected from surveys of Korean households, who used the personal robot in their home for 4 months before the survey, were used to verify the research model. This study showed that perceived usefulness and user satisfaction significantly positively impact the intention to use a personal robot service. In addition, perceived usefulness has a far more significant effect on the intention to use the service than does user satisfaction. The most influential factor was the motion quality of the ubiquitous personal robot, followed by tangible quality. Although the target of this study (the URC personal robot service) differs from that of our study (multi-robot service and integrated intelligent applications), both results are similar in that the motion quality which is similar to entertainment factor in our study affects user (customer) satisfaction.

[5] describes development of service robots evaluated in real restaurants. They developed two types of five restaurant service robots: two Order robots and three Server robots. To evaluate the robots, they had operated the five robots during crowded times (lunch and dinner), 4 hours a day, 5 days a week at the 5 branches of the restaurant in the Bangkok area. Since the stakeholders of the robots are the customers, general restaurant staff, and robot operator restaurant staff, the authors have observed the stakeholder's behaviors and have asked several general questions to evaluate the robots. Although the authors evaluated the robots from the viewpoints of robot production, robot operation, and project management, they did not evaluate the viewpoints of service quality and customer satisfaction in detail.

[7] presents an initial investigation of the effectiveness and service quality of a robot museum through the visitors experience. To evaluate the robot museum, three group of 57 visitors (aged 12–17) were invited to complete a questionnaire. The pre- and post-activity questionnaires consisted of the items about sociability, social attraction, and negative attitudes toward robots. In the post-activity questionnaire, the items for measuring the service quality, customer's loyalty, and satisfaction were added. To validate service quality, the authors used the items in [10], which were the measures based on SERVQUAL to evaluate service quality of the Hong Kong-based banking industry. Although they used the items based on SERVQUAL to evaluate the service quality of robot museum, most of the items were developed for evaluating staff and not for the robots. Items related to sociability, social attraction, and negative attitudes toward

the robots are mainly developed for robots. Some of the items are similar to our items related to the entertainment factor and interactivity.

## 6 Conclusion

In this paper, the evaluation of a multi-robot cafe from the view point of service quality has been reported. This evaluation was conducted through administering questionnaires to the guests in the multi-robot cafe that served as a practical application of PRINTEPS.

From the results, we found that the “entertainment factor” was an important factor in any objective variable and it was an indispensable factor in realizing the robot cafe service.

In terms of customer satisfaction, “assurance” and “responsiveness” were also key factors. In other words, to improve customer satisfaction, it is necessary to improve the service speed of the robot and enhance the robot’s ability to carefully carry out the promised service without delay.

“Customer satisfaction” strongly influenced “intention to revisit the robot cafe” and “intention to recommend the robot cafe to others”. Improving customer satisfaction is effective in increasing repeaters and new customers.

From this study, we found that PRINTEPS supports users in creating applications by considering the “entertainment factor”. To support the “entertainment factor”, we would like to extend PRINTEPS to support design thinking [17] and connect it with the workflows in PRINTEPS. We also would like to extend PRINTEPS to share the part of the workflows that are related to the “entertainment factor”. We believe that these extensions support users in easily applying their ideas related to entertainment to applications. We also found that the motion modules should be improved to make the movements faster. It is part of our future work to make the robots move faster while continuing to keep them from colliding into each other.

With the aim of making PRINTEPS an “AI for everybody” platform, we will add functions for end users to enable them to accumulate “knowledge of AI usage” in various application fields. We will also examine a method to evaluate the service quality of various integrated knowledge applications, and to identify key factors to improve customer satisfaction.

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